Experiment P12: Thermal Waves in a Metal Bar

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Abstract

This experiment investigates the propagation of thermal waves within an aluminum cylinder to characterize its dynamic thermal properties. By applying a periodic heat source at one end of the bar, sinusoidal temperature oscillations are induced and recorded at two fixed positions along its length. The attenuation and phase shift between these oscillations are analyzed using Orthogonal Distance Regression (ODR) to determine the thermal diffusivity and conductivity of the material. This method provides a direct measurement of the material's response to time-varying thermal gradients, allowing for a robust estimation of its transport coefficients under non-steady state conditions.

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2.1 Thermal Wave Analysis

The experimental temperature oscillations $\theta_1(t)$ and $\theta_2(t)$ were recorded and are shown in Figure 1. The waves exhibit a characteristic attenuation in amplitude and a phase shift as they propagate through the aluminum bar.

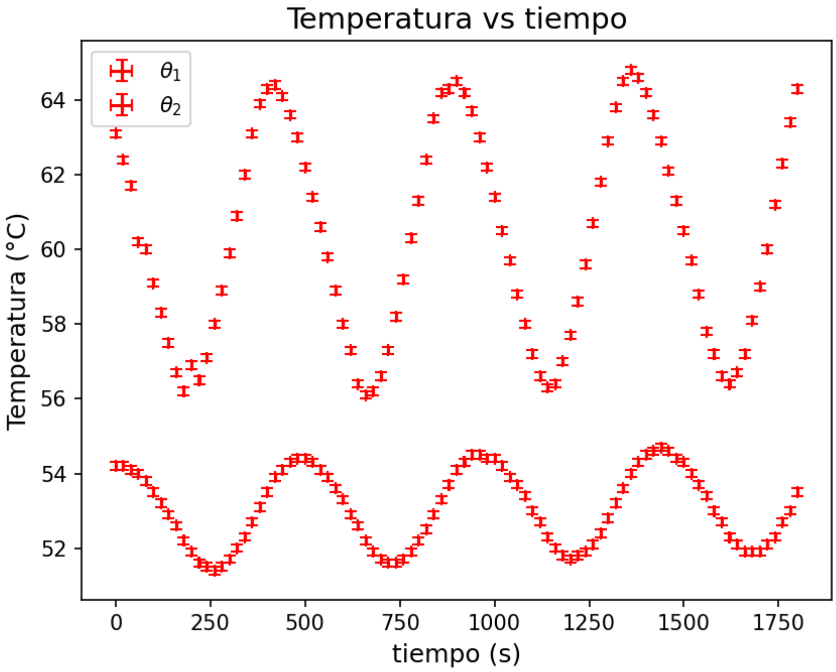

Figure 1: Temperature oscillations as a function of time at two sensor positions.

The data were fitted to a sinusoidal model using Orthogonal Distance Regression:

$$\theta(t) = A + \mathrm{Amp} \cdot \cos\left(\frac{2\pi}{\tau}t - \phi\right) \tag{1}$$

The results of the fit, following GUM standards for significant figures, are summarized in Table 1.

Table 1: Fitted parameters for the thermal oscillations at sensors 1 and 2.

Parameter	Sensor 1 (θ_1)	Sensor 2 (θ_2)
Mean Temp. A (°C)	53.110 ± 0.018	60.471 ± 0.031
Amplitude Amp (°C)	1.399 ± 0.025	-3.978 ± 0.044
Period τ (s)	475.8 ± 1.3	474.81 ± 0.72
Phase Shift ϕ (rad)	0.201 ± 0.037	2.486 ± 0.021

2.2 Derived Physical Parameters

Based on the attenuation and phase shift, the parameters m and h were calculated for the distance $d = (0.050 \pm 0.001)$ m. These parameters were then used to determine the experimental thermal diffusivity (K_{exp}) and thermal conductivity (λ_{exp}).

- $m = (20.91 \pm 0.60)\,\mathrm{m^{-1}}$
- $h = (45.7 \pm 1.3)\,\mathrm{m^{-1}}$
- $K_{exp} = (16.75 \pm 0.81)\,\mathrm{SI}$
- $\lambda_{exp} = (138.4 \pm 5.8)\,\mathrm{W\,m^{-1}\,K^{-1}}$

3 Discussion

4 Conclusions

5 Questions

A Experimental Data

The complete dataset recorded during the experiment is presented in Table 2.

Table 2: Full experimental measurements of temperature over time.

t (s)	u_t (s)	θ_1 (°C)	u_{θ_1} (°C)	θ_2 (°C)	u_{θ_2} (°C)
0.00	0.10	54.20	0.10	63.10	0.10
20.00	0.10	54.20	0.10	62.40	0.10
40.00	0.10	54.10	0.10	61.70	0.10
60.00	0.10	54.00	0.10	60.20	0.10
80.00	0.10	53.80	0.10	60.00	0.10
100.00	0.10	53.50	0.10	59.10	0.10
120.00	0.10	53.20	0.10	58.30	0.10
140.00	0.10	52.90	0.10	57.50	0.10
160.00	0.10	52.60	0.10	56.70	0.10
180.00	0.10	52.20	0.10	56.20	0.10
200.00	0.10	51.90	0.10	56.90	0.10
220.00	0.10	51.60	0.10	56.50	0.10
240.00	0.10	51.50	0.10	57.10	0.10
260.00	0.10	51.40	0.10	58.00	0.10
280.00	0.10	51.50	0.10	58.90	0.10
300.00	0.10	51.70	0.10	59.90	0.10
320.00	0.10	52.00	0.10	60.90	0.10
340.00	0.10	52.30	0.10	62.00	0.10
360.00	0.10	52.70	0.10	63.10	0.10
380.00	0.10	53.10	0.10	63.90	0.10
400.00	0.10	53.50	0.10	64.30	0.10
420.00	0.10	53.90	0.10	64.40	0.10
440.00	0.10	54.10	0.10	64.10	0.10

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Table 2 – continued from previous page

t (s)	u_t (s)	θ_1 (°C)	u_{θ_1} (°C)	θ_2 (°C)	u_{θ_2} (°C)
460.00	0.10	54.30	0.10	63.60	0.10
480.00	0.10	54.40	0.10	63.00	0.10
500.00	0.10	54.40	0.10	62.20	0.10
520.00	0.10	54.30	0.10	61.40	0.10
540.00	0.10	54.10	0.10	60.60	0.10
560.00	0.10	53.90	0.10	59.80	0.10
580.00	0.10	53.60	0.10	58.90	0.10
600.00	0.10	53.30	0.10	58.00	0.10
620.00	0.10	52.90	0.10	57.30	0.10
640.00	0.10	52.60	0.10	56.40	0.10
660.00	0.10	52.20	0.10	56.10	0.10
680.00	0.10	51.90	0.10	56.20	0.10
700.00	0.10	51.70	0.10	56.60	0.10
720.00	0.10	51.60	0.10	57.30	0.10
740.00	0.10	51.60	0.10	58.20	0.10
760.00	0.10	51.70	0.10	59.20	0.10
780.00	0.10	51.90	0.10	60.30	0.10
800.00	0.10	52.20	0.10	61.30	0.10
820.00	0.10	52.50	0.10	62.40	0.10
840.00	0.10	52.90	0.10	63.50	0.10
860.00	0.10	53.30	0.10	64.20	0.10
880.00	0.10	53.70	0.10	64.30	0.10
900.00	0.10	54.10	0.10	64.50	0.10
920.00	0.10	54.30	0.10	64.20	0.10
940.00	0.10	54.50	0.10	63.70	0.10
960.00	0.10	54.50	0.10	63.00	0.10
980.00	0.10	54.40	0.10	62.20	0.10
1000.00	0.10	54.40	0.10	61.40	0.10
1020.00	0.10	54.20	0.10	60.50	0.10
1040.00	0.10	53.90	0.10	59.70	0.10
1060.00	0.10	53.70	0.10	58.80	0.10
1080.00	0.10	53.40	0.10	58.00	0.10
1100.00	0.10	53.00	0.10	57.20	0.10
1120.00	0.10	52.70	0.10	56.60	0.10
1140.00	0.10	52.30	0.10	56.30	0.10
1160.00	0.10	52.00	0.10	56.40	0.10
1180.00	0.10	51.80	0.10	57.00	0.10
1200.00	0.10	51.70	0.10	57.70	0.10
1220.00	0.10	51.80	0.10	58.60	0.10

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Table 2 – continued from previous page

t (s)	u_t (s)	θ_1 (°C)	u_{θ_1} (°C)	θ_2 (°C)	u_{θ_2} (°C)
1240.00	0.10	51.90	0.10	59.60	0.10
1260.00	0.10	52.10	0.10	60.70	0.10
1280.00	0.10	52.40	0.10	61.80	0.10
1300.00	0.10	52.80	0.10	62.90	0.10
1320.00	0.10	53.20	0.10	63.80	0.10
1340.00	0.10	53.60	0.10	64.50	0.10
1360.00	0.10	54.00	0.10	64.80	0.10
1380.00	0.10	54.30	0.10	64.60	0.10
1400.00	0.10	54.50	0.10	64.20	0.10
1420.00	0.10	54.60	0.10	63.60	0.10
1440.00	0.10	54.70	0.10	62.90	0.10
1460.00	0.10	54.60	0.10	62.10	0.10
1480.00	0.10	54.40	0.10	61.30	0.10
1500.00	0.10	54.30	0.10	60.50	0.10
1520.00	0.10	54.00	0.10	59.70	0.10
1540.00	0.10	53.70	0.10	58.80	0.10
1560.00	0.10	53.40	0.10	57.80	0.10
1580.00	0.10	53.00	0.10	57.20	0.10
1600.00	0.10	52.70	0.10	56.60	0.10
1620.00	0.10	52.30	0.10	56.40	0.10
1640.00	0.10	52.10	0.10	56.70	0.10
1660.00	0.10	51.90	0.10	57.20	0.10
1680.00	0.10	51.90	0.10	58.10	0.10
1700.00	0.10	51.90	0.10	59.00	0.10
1720.00	0.10	52.10	0.10	60.00	0.10
1740.00	0.10	52.30	0.10	61.20	0.10
1760.00	0.10	52.70	0.10	62.30	0.10
1780.00	0.10	53.00	0.10	63.40	0.10
1800.00	0.10	53.50	0.10	64.30	0.10